SAL Institute of Technology & Engineering Research

DEPARTMENT OF COMPUTER ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Mid Examination Syllabus **Sem - VII**
Subject Name: Machine Learning **Subject Code : 3170724**

Unit	Topics and Sub-Topics
I - Introduction to Machine Learning:	Overview of Human Learning and Machine Learning, Types of Machine Learning, Applications of Machine Learning, Tools and Technology for Machine Learning.
II - Preparing to Model:	Machine Learning activities, Types of data in Machine Learning, Structures of data, Data quality and remediation, Data Pre-Processing: Dimensionality reduction, Feature subset selection.
III - Modelling and Evaluation:	Selecting a Model: Predictive/Descriptive, Training a Model for supervised learning, model representation and interpretability, Evaluating performance of a model, Improving performance of a model.
IV - Basics of Feature Engineering:	Feature and Feature Engineering, Feature transformation: Construction and extraction, Feature subset selection : Issues in high-dimensional data, key drivers, measure and overall process.
V - Overview of Probability:	Statistical tools in Machine Learning, Concepts of probability, Random variables, Discrete distributions, Continuous distributions, Multiple random variables, Central limit theorem, Sampling distributions, Hypothesis testing, Monte Carlo Approximation.

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